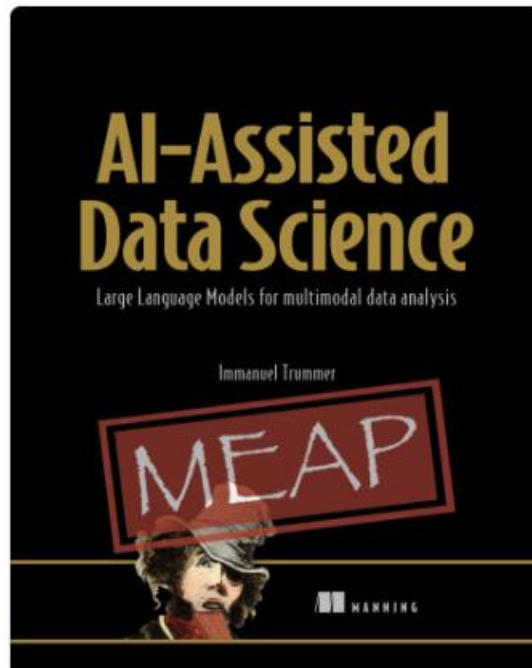


Trummer/AI-Assisted Data Science



about the author



Immanuel Trummer is an assistant professor for computer science at Cornell University and Head of the Cornell Database Group (<https://www.cs.cornell.edu/research/datasystems>).

Immanuel's papers have been selected for "Best of VLDB", "Best of SIGMOD", for the ACM SIGMOD Research Highlight Award, and for publication in CACM as CACM Research Highlight. Immanuel's online course on data management has reached close to a million views on YouTube. Over the past few years, his group has published extensively on projects that apply large language models in the context of data science. This book is a largely extended version of a series of tutorials, introducing data scientists to large language models, given by Immanuel at premier conferences in data management such as VLDB and BTW.

about the book

AI-Assisted Data Science teaches you to use a new generation of AI assistants and Large Language Models (LLMs) to simplify and accelerate common data science tasks. Cornell professor and long-time LLM advocate Immanuel Trummer reveals techniques he's pioneered for getting the most out of LLMs in data science, including model selection and specialization, techniques for tuning parameters, and reliable prompt templates.

You'll start with an in-depth exploration of how LLMs work. Then, you'll dive into no-code data analysis with LLMs, creating custom operators with the OpenAI Python API, and building complex data analysis pipelines with the cutting edge LangChain framework.

Takeaways:

- Analyze text, tables, images, and audio files
- Extract information from multi-modal data lakes
- Classify, cluster, transform, and query multimodal data
- Build natural language query interfaces over structured data sources
- Use LangChain to build complex data analysis pipelines
- Prompt engineering and model configuration

[minimally qualified reader](#)

For data scientists, data analysts, and others who are interested in making their work easier through the use of artificial intelligence techniques. Readers should have a basic understanding of the Python programming language.

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